HACKFRONT INDIA 2026 · AI AND DEVELOPER TOOLS (T-1)

WhyCode

Why is this code like this?
Ask its own history.

Team FORTEC Ronit Shrimankar (lead) · Anshul Kamthe · Udita Singh

Built on **django/django**

01 · THE PROBLEM

Decisions nobody can explain any more

- **Codebases accumulate decisions** whose reasons are forgotten.
- **New developers guess** why a piece of code exists.
- **A wrong guess breaks things** the original author fixed on purpose.
- **The reasons are scattered** across commits, pull requests and tickets.

Singh

02 · THE SOLUTION

Ask the history, get a cited answer

Ingest

A public repo's history becomes a graph.

Ask

Plain-language *why* questions.

Evidence

Every answer carries its chain + a confidence score.

Refuse

No evidence? It refuses. Never invents.

03 · HOW IT WORKS

From history to answer

1. **Ingest** — `gh` CLI for commits & PRs, plus Trac tickets and Django Enhancement Proposals. Django uses Trac, not GitHub Issues.
2. **Graph** — entities: commit, PR, ticket, file, person, design doc. Edges: authored, fixes, references, touches, supersedes, mentions.
3. **Ask** — evidence chain, supersession flags, refusal when evidence is absent.
4. **Runs anywhere** — bundled SQLite database, no credentials, no paid APIs.

public history.

04 · DEMO — A REAL ANSWER

ignore_warnings, explained

```
$ python -m whycode.cli --question "Why does
  ignore_warnings match substrings by default?"
refused: false   confidence: 0.99
Best match: Fixed #37089 — Made ignore_warnings()
match substrings by default. (commit)
Evidence:
→ fixes Trac ticket #37089
→ authored by HelmiDev03
```

Repo: github.co
Built during the hack

05 · DEMO — THE TEMPLATE-TAGS OPTION

Multiline tags, traced

```
$ python -m whycode.cli --question "Why is there
  a transitional option to disable multiline
  template tags?"

refused: false   confidence: 0.86

Best match: Refs #35899 — Added allow_multiline_tags
transitional template option. (commit)

Evidence:

→ references Trac ticket #35899

→ authored by boxed
```

06 · THE REFUSAL

When it doesn't know, it says so

```
$ python -m whycode.cli --question "Why does
  Django use a microservices architecture?"
refused: true   confidence: 0.0
No evidence in the graph answers this.
Refusing to invent an answer.
```

> An earlier version answered this from an unrelated docs commit. The absent-term rule + a regression test fixed that for good.

07 · ENGINEERING

Runnable, tested, no magic

200 commits	**100** pull requests	**100** Trac tickets	**161** files
90 people	**27** design docs	**641** edges	

Python standard library only · SQLite graph · **13/13 tests** · bundled database runs with **zero credentials**.

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WHYCODE — BY FORTEC

Questions welcome.

```
$ python -m whycode.cli --db data/whycode.db
  --question "Why ...?"
```

Repo: **github.com/eres0045/hackfront-fortec**

Built during the hackFront league timeline, from Django's real, public history.